Sahaj Bhadja

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Summary

Forward Deployed Software Engineer with 2+ years of experience building **production Python systems**, **AI workflows**, and reliable customer-facing platforms across cloud infrastructure.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Database Systems, Analysis of Algorithms, Web Technologies

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Software Engineering, Object-Oriented Programming, Operating Systems

Experience

USC Dornsife **September 2025 – Present**
Forward Deployed Software Engineer *Los Angeles, CA*

- Tripled throughput across 1K+ AI jobs by restructuring **fault-tolerant orchestration** with explicit job-state transitions, failure isolation, recoverable background processing, and observable concurrent workflows under sustained production load reliably.
- Achieved 99%+ reliability across 5K+ monthly jobs through **production data pipelines**, implementing incremental processing, validation, reconciliation, error handling, cloud storage, CI/CD releases, and stable production observability reliably.

Grad8 **June 2025 – August 2025**
Software Engineer Intern *Los Angeles, CA*

- Reduced feedback turnaround by 60% through a **production Python platform** supporting 150+ instructors and 10K+ daily requests with REST APIs, AWS RDS, secure ingestion, and CI/CD workflows.
- Processed 1K+ files per batch using **Python multithreading**, validating heterogeneous inputs while isolating failures and preserving responsive customer workflows during concurrent model execution and production traffic reliably.

USC AD Labs **January 2025 – May 2025**
ML Research Assistant *Los Angeles, CA*

- Improved long-horizon trajectory planning by 18% in simulation by building a **Decision Transformer pipeline** over 100K+ autonomous-driving logs with offline reinforcement learning and structured evaluation workflows reliably.
- Improved closed-loop planning stability by 15% through a **spatial-temporal encoder** combining HD-map geometry, ego trajectories, nearby-agent context, multi-head attention, and distance-aware interaction modeling across complex simulation scenarios.

Aasma Technology Solutions **August 2023 – July 2024**
Software Developer Engineer - (SecureSync) *India*

- Improved API response times by 30% through **maintainable microservices**, indexing, caching, pagination, and filtered payloads while troubleshooting integrations across Windows, Linux, and macOS production environments reliably at scale.
- Increased pipeline efficiency by 68% through **AWS Glue and S3 automation**, transforming structured data while reducing storage costs by 15% and supporting repeatable cloud delivery workflows at scale.

Aasma Technology Solutions **May 2023 – August 2023**
Full Stack Developer Intern - (VibrantChem) *India*

- Synchronized 2M+ records through an eight-service **customer operations platform** covering authentication, resources, customers, billing, leads, permissions, and notifications with Node.js, WebSockets, and Azure SQL across locations reliably.
- Reduced resource wastage by 42% through **real-time workflow synchronization**, integrating automated tests, Docker/Jenkins deployments, Azure Monitor, Sentry, and Kibana across modular production services and delivery workflows reliably.

Skills

Languages: Python, TypeScript, JavaScript, Java, SQL, C++, Bash
AI & Agents: LLMs, Retrieval-Augmented Generation, Vector Databases, FAISS, Pinecone, Transformers, PyTorch, Model Evaluation
Backend & Frontend: FastAPI, Flask, Node.js, React, REST APIs, WebSockets, Microservices, API Design
Data & Cloud: PostgreSQL, AWS RDS, AWS S3, AWS Glue, GCP, Azure, Docker, Kubernetes
Production Engineering: Multithreading, Distributed Systems, Caching, Data Validation, Error Handling, Observability, Performance Optimization, CI/CD, Jenkins

Projects

Multimodal Service Operations Agent - Design Concept

- Designed a **multimodal AI agent** concept using FastAPI, retrieval-augmented generation, voice/text tool calls, structured workflow state, human escalation, latency budgets, evaluation datasets, and traceable customer outcome metrics.

Customer Deployment Evaluation Harness - Design Concept

- Designed a **deployment evaluation harness** concept supporting replayable conversations, model comparisons, prompt and retrieval variants, success metrics, failure taxonomies, A/B assignments, observability, and automated customer-specific regression gates.

Big-Brain | Live

- Built a **retrieval-augmented generation platform** using LLaMA-2, FAISS, Pinecone, PostgreSQL, GCP, and REST APIs, sustaining 95% query accuracy while enabling real-time document analysis reliably for 200+ users.